MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

TEST POINT, ISOLATED, MISCELLANEOUS, KEY CODE FZJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic isolated test point module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

MIL-P-46112 - Plastic Sheet and Strip, Polyimide.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

NAVORD Drawing 10001-2316885 - Eyelet, Metallic.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.039 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.4.1 Life test. When tested in accordance with 4.5.3, the resistance shall not exceed ±1.5 percent of the 25°C post operating temperature dc resistance.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FZJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The fin extraction holes are not required (see figure 2).

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified on figure 3, and the module shall be marked with the military part number M28787/211-1.

3.5.9 Special mechanical configuration. Figures 2 and 3 specify the special mechanical configuration.

3.5.10 Insulator. The insulator shall be 0.007 ±0.0005 inch thick film in accordance with MIL-P-46112, type I, or equivalent.

3.5.11 Eyelets. Eyelets are noninsulated and are located on 0.225 inch centers (see figure 2). These eyelets are intended for use with spring hook type slip-on probe tips similar to Tektronic slip-on 013-0090-00 and Hewlett Packard H/P 10229A, or equivalent. Eyelets shall be in accordance with NAVORD Drawing 10001-2316885.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

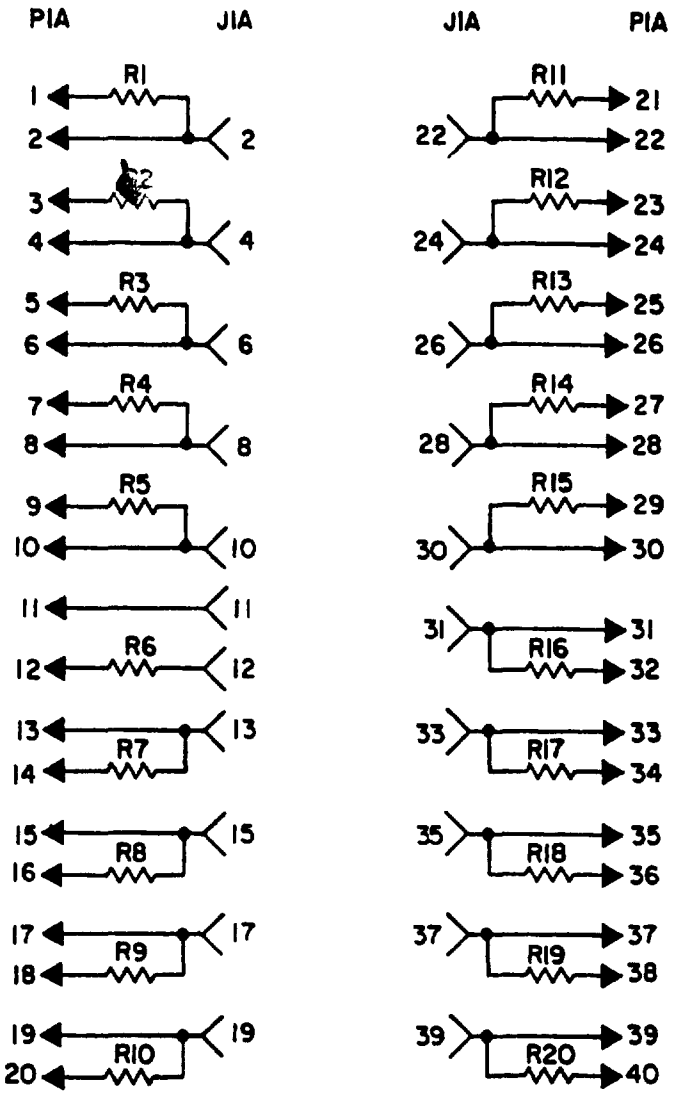
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

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NOTE: R1-R20 = 10 kΩ ±1%, 1/8 W.

* FIGURE 1. FZJ module function diagram, isolated test point.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/,2/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A5	Isolation, pin	II	S	10	∞	1	∞	--	--	--	--	MΩ
B	Isolation, frame	II	X	10	∞	1	∞	--	--	--	--	MΩ
C1-C20	DC resistance	III	X	9.9	10.1	See para 3.4.1	See para 3.4.1	9.825	10.175	9.3	10.7	kΩ
D1-D20	Continuity	IV	X	0	1	0	1	--	--	--	--	Ω

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

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* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through D shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through D in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test condition C shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times the module shall be nonoperating.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and while still at the maximum class temperature, electrical test C shall be performed. The resistance values measured shall conform to the 0°C to 60°C end-of-life requirements of table I. Following the life test, test condition C shall be performed using the requirements of 3.4.1.

4.5.4 Mechanical tests. Fin torque test and fin cantilever test are not required.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/211-1 supersedes part number M28787/211. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	
Resistor (each)	5 mW
Module	100 mW
Maximum applied voltage	±30 V

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6.5 Module function. This module provides up to 20 test points plus chassis frame connection. The module contains twenty 10 kΩ ±1 percent, 1/8 W resistors which may be pin programmed to provide resistor isolation for test points. Alternate pin programming may be used to remove resistor isolation for all but one test point. The module may also be used to provide resistive loads but because of the absence of a cooling fin, the total module power dissipation shall not exceed 0.1 W. The eyelet test points are intended for use with spring hook type test probes.

6.5.1 Module applications.

6.5.1.1 Voltages in excess of 30 V may result in a personnel hazard and should not be used with this module.

6.5.1.2 All voltages available at test points should be current limited either at the source or with 10 kΩ resistors available in the module.

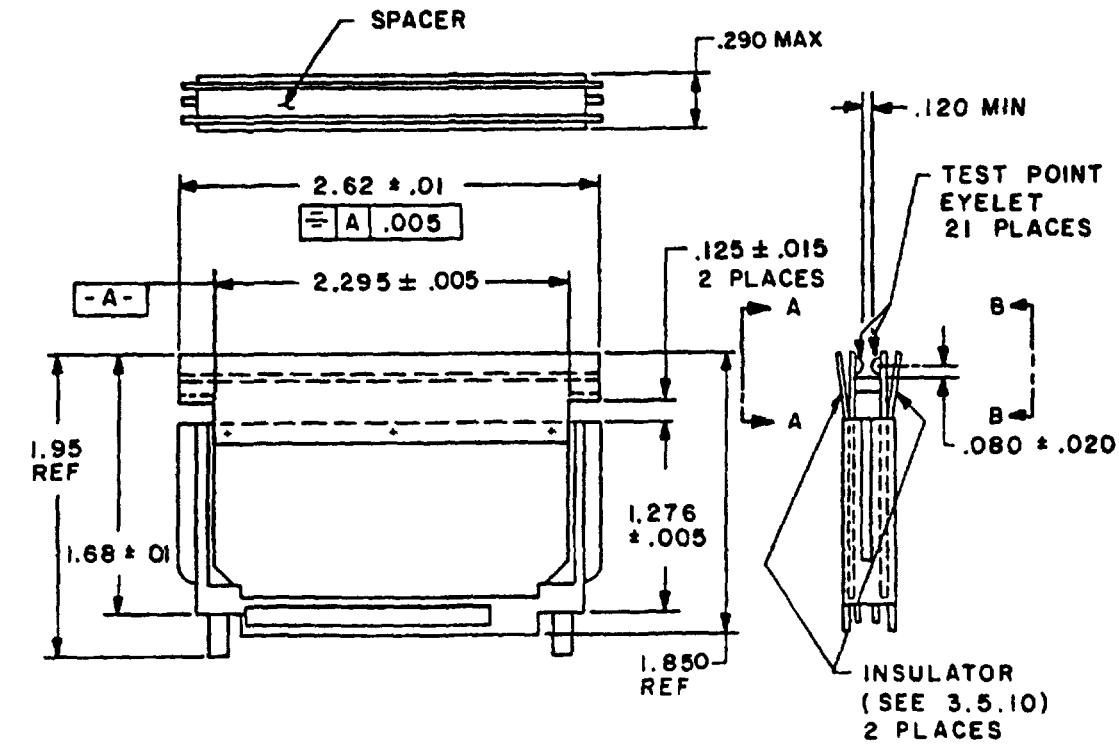
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

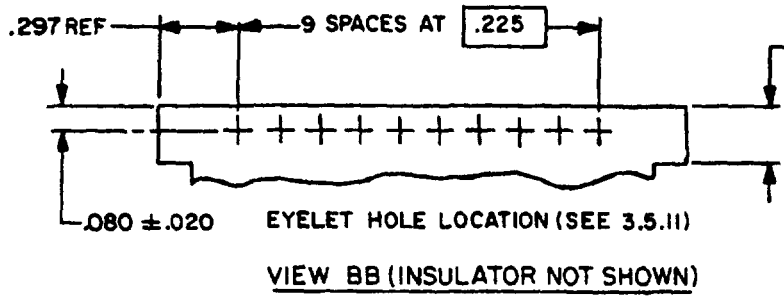
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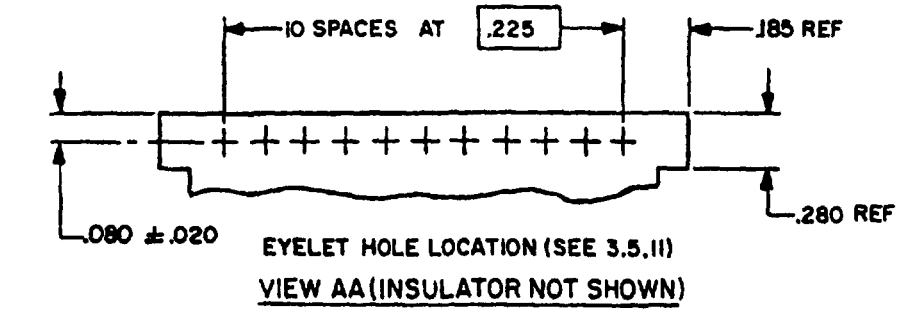
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VIEW BB (INSULATOR NOT SHOWN)

INCHES	MM	INCHES	MM
.005	.13	.280	7.11
.010	.25	.290	7.37
.01	.3	.300	7.62
.020	.51	1.276	32.41
.080	2.03	1.68	42.7
.090	2.29	1.850	46.99
.120	3.05	1.95	49.5
.185	4.69	2.29	58.2
.225	5.72	2.62	66.6



VIEW AA (INSULATOR NOT SHOWN)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Module configuration requirements.

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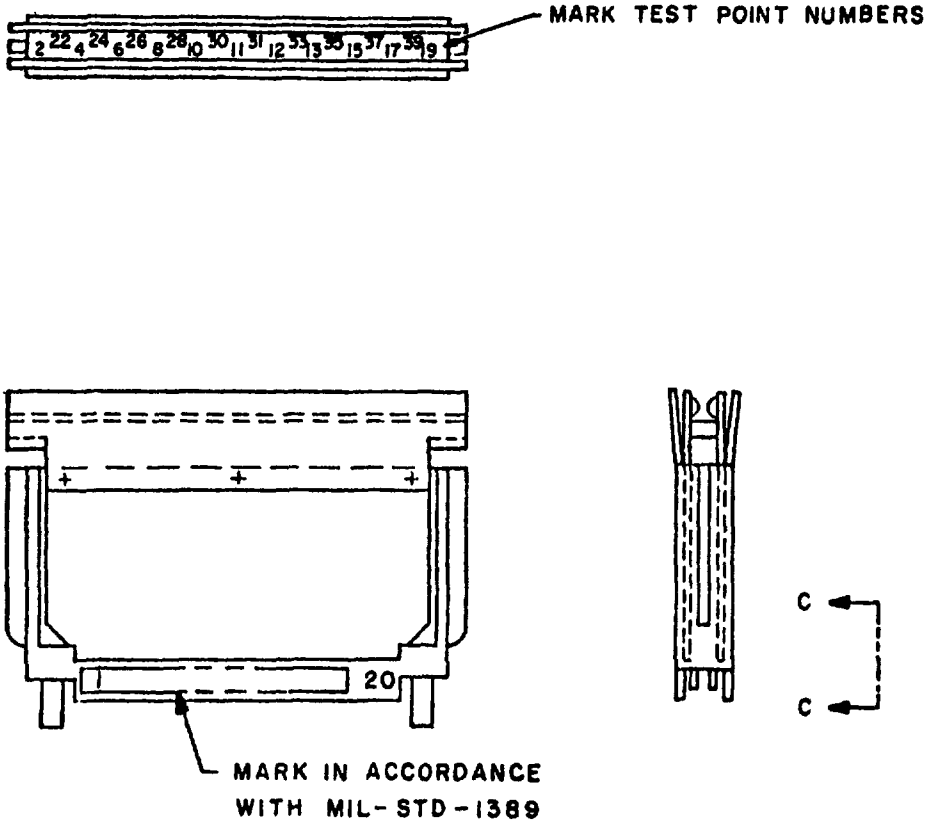
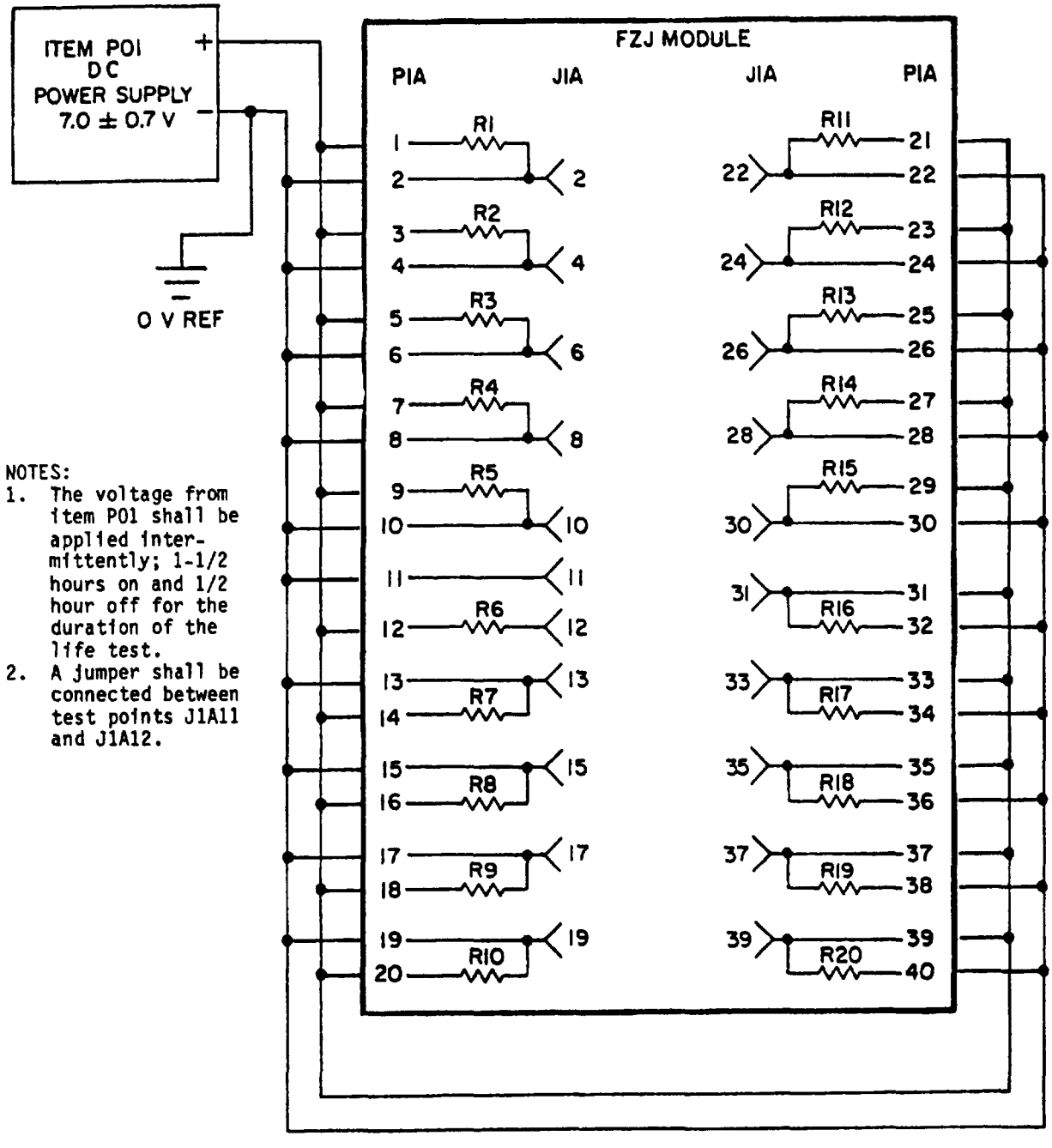


FIGURE 3. Module marking requirements.

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NOTES:
1. The voltage from item P01 shall be applied intermittently; 1-1/2 hours on and 1/2 hour off for the duration of the life test.
2. A jumper shall be connected between test points J1A11 and J1A12.

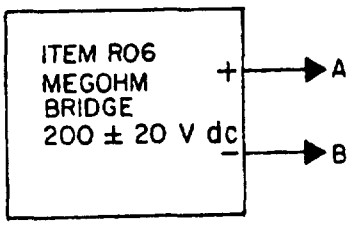
FIGURE 4. Life test and component temperature test circuit.

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* TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, pin	1,2,5,6,9,10, 12,15,16,19, 20,21,22,25, 26,29,30,33, 34,37,38	3,4,7,8,11,13, 14,17,18,23, 24,27,28,31, 32,35,36,39, 40	10	∞	MΩ	5
A2	"	1,2,3,4,9,10, 11,15,16,17, 18,21,22,23, 24,29,30,31, 32,37,38,39, 40	5,6,7,8,12,13, 14,19,20,25, 26,27,28,33, 34,35,36	"	"	"	"
A3	"	1,2,3,4,5,6,7, 8,15,16,17,18, 19,20,21,22, 31,32,33,34, 35,36,37,38	9,10,11,12,13, 14,23,24,25, 26,27,28,29, 30,39,40	"	"	"	"
A4	"	1,2,3,4,5,6,7, 8,9,10,21,22, 23,24,25,26, 27,28,29,30	11,12,13,14, 15,16,17,18, 19,20,31,32, 33,34,35,36, 37,38,39,40	"	"	"	"
A5	"	1,2,3,4,5,6,7, 8,9,10,11,12, 13,14,15,16, 17,18,19,20	21,22,23,24, 25,26,27,28, 29,30,31,32, 33,34,35,36, 37,38,39,40	"	"	"	"
B	Isolation, frame	All pins	Frame	"	"	"	"

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition B, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

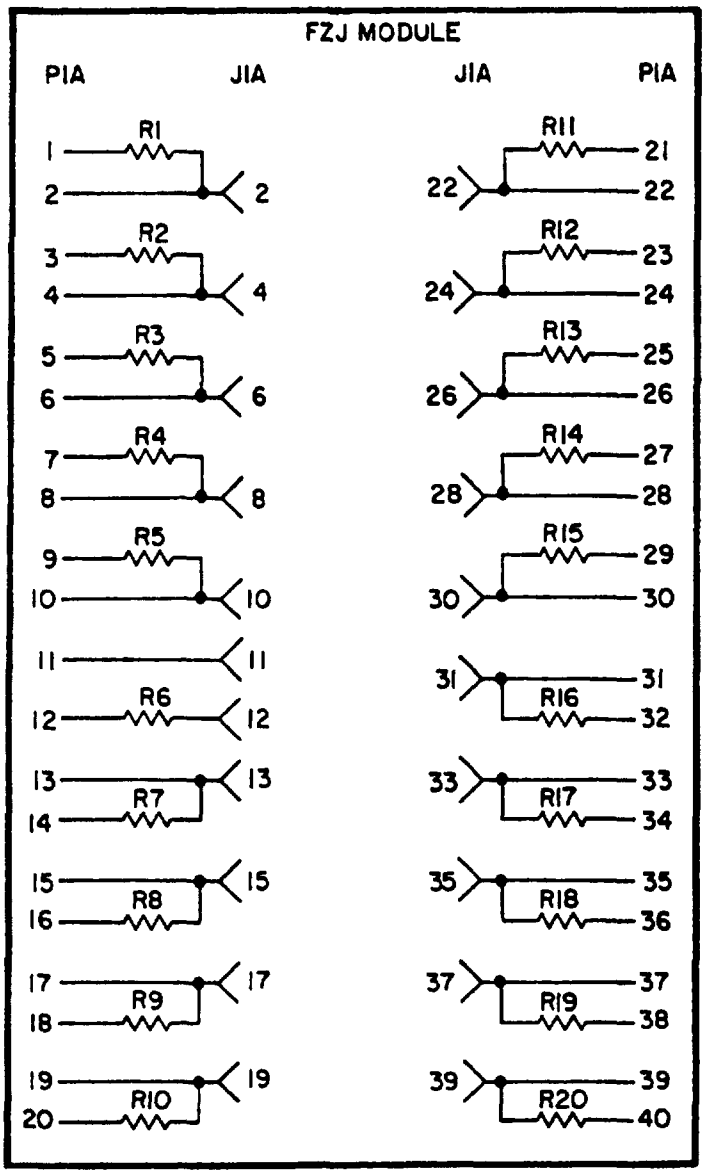


FIGURE 5. Isolation test circuit for table II.

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TABLE III. DC resistance test.

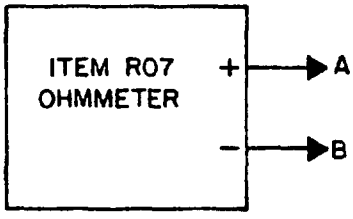
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	DC resistance	1	2	9.9	10.1	kΩ	6
C2	"	3	4	"	"	"	"
C3	"	5	6	"	"	"	"
C4	"	7	8	"	"	"	"
C5	"	9	10	"	"	"	"
C6	"	12	11	"	"	"	"
C7	"	14	13	"	"	"	"
C8	"	16	15	"	"	"	"
C9	"	18	17	"	"	"	"
C10	"	20	19	"	"	"	"
C11	"	21	22	"	"	"	"
C12	"	23	24	"	"	"	"
C13	"	25	26	"	"	"	"
C14	"	27	28	"	"	"	"
C15	"	29	30	"	"	"	"
C16	"	32	31	"	"	"	"
C17	"	34	33	"	"	"	"
C18	"	36	35	"	"	"	"
C19	"	38	37	"	"	"	"
C20	"	40	39	"	"	"	"

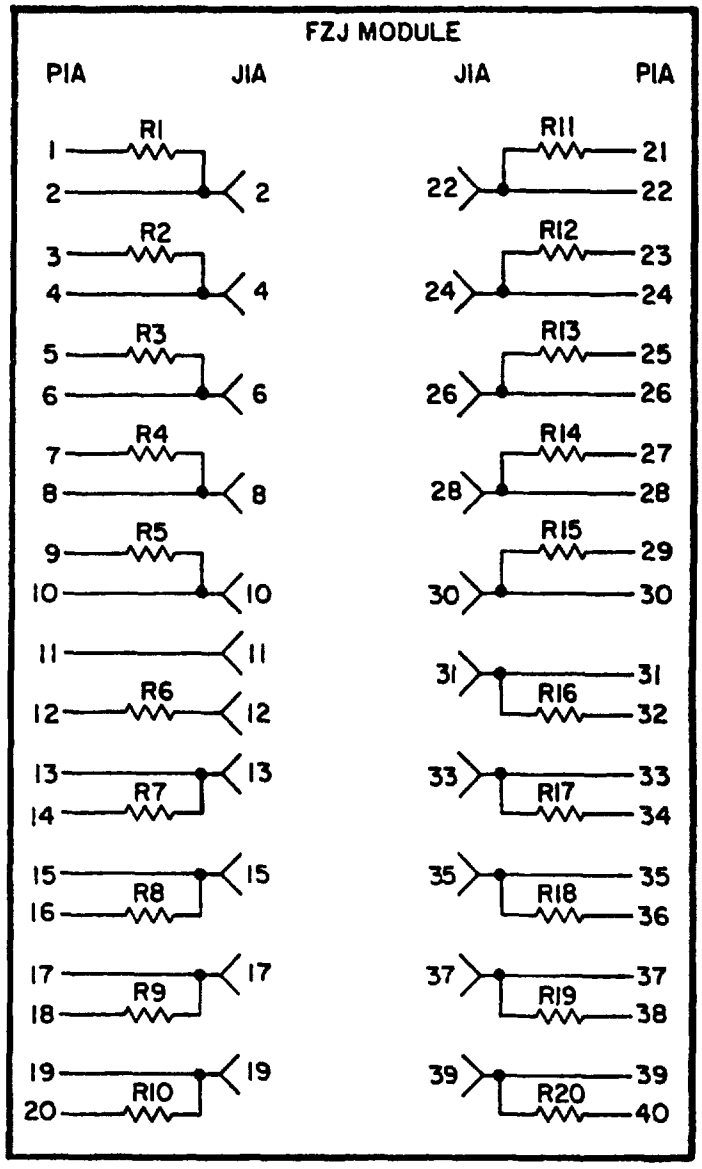
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TABLE IV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	Continuity	2	J1A2	0	1	Ω	6
D2	"	4	J1A4	"	"	"	"
D3	"	6	J1A6	"	"	"	"
D4	"	8	J1A8	"	"	"	"
D5	"	10	J1A10	"	"	"	"
D6	"	11	J1A11	"	"	"	"
D7	"	13	J1A13	"	"	"	"
D8	"	15	J1A15	"	"	"	"
D9	"	17	J1A17	"	"	"	"
D10	"	19	J1A19	"	"	"	"
D11	"	22	J1A22	"	"	"	"
D12	"	24	J1A24	"	"	"	"
D13	"	26	J1A26	"	"	"	"
D14	"	28	J1A28	"	"	"	"
D15	"	30	J1A30	"	"	"	"
D16	"	31	J1A31	"	"	"	"
D17	"	33	J1A33	"	"	"	"
D18	"	35	J1A35	"	"	"	"
D19	"	37	J1A37	"	"	"	"
D20	"	39	J1A39	"	"	"	"

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NOTE:
For test condition C6, a jumper shall be connected between test points J1A11 and J1A12.

* FIGURE 6. DC resistance and continuity test circuit for tables III and IV.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/211A(EC)	Modules, Standard Electronic Test Point, Isolated, Miscellaneous, Key Code FZJ

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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5 PROBLEM AREAS

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